EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Once a cached value is stored it is always correct.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.



RHS COMPUTING
OCR H446 · LESSON 239 · DRAFT

DO NOW — RECALL

# Thinking ahead: inputs, caching and reuse

Name: _____ Class: _____ Date: _____

**1. Why omit cover colour from a simple book-loan model?**

**2. What is a reusable subroutine?**

**3. Calculate 7×7.**

**4. Why is caching squares safer than caching changing attendance data?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Cover colour does not determine availability.

### 2. Model answer

A function/procedure with a clear interface usable in multiple places.

### 3. Model answer

49.

### 4. Model answer

Squares of the same integer do not change; attendance does.

### Vocabulary — know these terms

input — Data passed into a computation.

output — Result of a computation.

precondition — Condition required before a process can run correctly.

cache hit — Requested value found in storage.

cache invalidation — Removing or refreshing outdated stored results.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Give three advantages/disadvantages of caching changing school data and a way to avoid stale values.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

For SquareCached(6), identify input, output and two preconditions or assumptions.

### Question 2

Trace two calls of SquareCached(7) starting with an empty cache.





NEW LEARNING

# Read, annotate and explain

### Define the contract

The input is an integer, and the output is its square; overflow is a possible precondition concern.

### Trace repeated work

First call for 5 calculates and caches 25; second returns the stored result.

### Evaluate caching

Saves repeated calculation but costs memory. For changing data, cached values can become out of date.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**A cached ticket price for zone A is £3.40. The same zone is requested twice.**

### Predict first

The key A is already stored, so the lookup uses the cache.

### Model and check

If pricing changes, the cached price must be updated or expired.





WE DO — GUIDED PRACTICE

# Try → Discuss → Improve